

SAI RAM VARMA BUDHARAJU

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SUMMARY

Software Engineer with 5+ years designing and shipping scalable backend systems, async processing pipelines, and event-driven architectures across cloud-native environments on AWS and Kubernetes. Proficient in **Python (FastAPI, Django, Flask), Java (Spring Boot), PostgreSQL, and Kafka**, with a track record of integrating ML inference and RAG pipelines into high-volume production systems handling 300K+ daily records. Open-source author of **pgsemantic**, a **python** library that adds semantic search to any Postgres database.

PROFESSIONAL EXPERIENCE

Software Engineer

Feb 2026 – Present

UBER

USA

- Built real-time event ingestion pipelines using Kafka and Python to process ride and support telemetry, achieving sub 100ms data freshness for dashboards used by 50+ support agents.
- Redesigned the query layer across MySQL and MongoDB through composite indexing and schema tuning, reducing p95 retrieval latency from 3.2s to 1.4s for customer support backend services.
- Architected REST APIs in Node.js and Express with role-based access control, supporting secure data exchange across customer support microservices and cutting auth-related defects by 3 per release.
- Integrated ML-based ticket classification into the support routing pipeline using FastAPI and Scikit-learn, automating triage for 12K+ weekly tickets and reducing average first-response time by 35%.
- Architected CI/CD pipelines using GitHub Actions and AWS CodePipeline for containerized services on EKS, cutting deployment lead-time by 6 hours per sprint and eliminating 3 manual release steps.

Software Developer

Mar 2025 – Dec 2025

Northeastern University

Washington, USA

- Built AI-powered grant eligibility workflows using RAG with OpenAI GPT and LangChain, served via FastAPI with Scikit-learn classification models, processing 300+ records per cycle with 92% accuracy.
- Implemented event-driven enrollment and grant processing pipelines using Apache Kafka and Spring Boot on AWS EKS, designing topic partitioning that sustained 5K+ events/hour with zero message loss.
- Refactored 7 legacy auth modules to OAuth 2.0 and JWT on Kubernetes, reducing production defects from 8 to 2 per release cycle and cutting auth-related incident response by 60%.
- Deployed Prometheus and Grafana monitoring across 4 microservices, enabling proactive alerting that reduced undetected service degradations by 70% and cut mean time to recovery from 45 to 12 minutes.

Software Developer

May 2024 – Feb 2025

University of Florida

Florida, USA

- Developed REST APIs using FastAPI and PostgreSQL for research data access, reducing average query response time by 35% through indexing and query plan optimization.
- Automated recurring ETL workflows using Apache Airflow and Python, cutting report generation time from 6 hours to 2 hours weekly across student and administrative datasets.
- Deployed microservices on AWS EKS using Docker and Kubernetes, configuring health checks and autoscaling to maintain 99.5% uptime across internal research portals.
- Integrated OAuth 2.0 authentication and set up GitHub Actions CI/CD pipelines, enabling automated testing and zero-touch deployments across dev and staging environments.

Software Developer/Assistant Systems Engineer

Oct 2020 – Apr 2022

Tata Consultancy Services

India

- Built backend transaction services using Flask, SQLAlchemy, and Oracle Database integrating 4 financial source systems, processing 300K+ daily records with 99.9% settlement accuracy for production reporting.
- Orchestrated 130+ daily settlement jobs using Celery and RabbitMQ, reducing batch runtime from 2 hours to 55 minutes and improving SLA adherence by 30%.
- Built an XGBoost risk scoring pipeline processing 8K+ daily financial transactions, flagging high-risk entries with 87% precision and reducing manual review workload by 45% for the operations team.
- Engineered PySpark and Pandas data transformation pipelines consolidating 800K+ weekly records from 6 upstream systems, cutting reconciliation processing time by 50%.
- Automated monthly data ingestion using Azure Functions, eliminating 3 full days of manual processing per cycle and improving dataset delivery reliability across 4 client accounts.

Software Developer

May 2019 – Sep 2020

Airbnb

India

- Connected booking workflows to ML recommendation models by building a Spring Boot to Flask inference shim on AWS EC2, handling 4.5K calls/day and improving p95 latency from 1.2s to 0.8s.
- Orchestrated nightly Airflow pipelines syncing PostgreSQL to Snowflake, preparing 2–3 pricing datasets daily and eliminating manual analyst data pulls entirely.

- Automated CI/CD pipelines with Jenkins, reducing manual deployment efforts by 40%, and implemented containerization with Docker and Kubernetes for zero-downtime deployments.
- Expanded integration test coverage by 40+ cases through schema contract alignment with PM, DS, and QA partners, reducing cross-team integration bugs by 30%.
- Prototyped a semantic search engine using TF-IDF and Elasticsearch, indexing 1.2K property descriptions and reducing onboarding lookup time to under 450ms (p50).

PROJECTS

Pgsemantic- Zero-config semantic search for PostgreSQL · Published on PyPI

Python+PostgreSQL (pgvector)+FastAPI

- Built a zero-config semantic search layer for PostgreSQL, supporting 5 embedding providers across 384–3,072 dimensions with HNSW indexing and automatic embedding sync via database triggers — no schema migration required.
- Shipped a 7-tool MCP server in FastAPI exposing semantic and hybrid search to AI agents (Claude, Cursor) over stdio and SSE, with CSRF protection, SQL injection prevention via Pydantic validation, and rate limiting at 120 req/min.
- Published to PyPI with a full CLI and web UI enabling setup in under 60 seconds, backed by an automated test suite of 144 tests covering search accuracy, embedding sync, and API security.

Chorus - Federated LoRA fine-tuning framework · Published on PyPI ·

Distributed Systems + Federated Learning + AI Infrastructure

- Implemented FedEx-LoRA (ACL/ICLR 2025) in Python — an exact federated LoRA aggregation algorithm using SVD residual folding that corrects the mathematical flaw in standard FedAvg for LoRA adapters, published on PyPI as chorus-fl.
- Built a FastAPI aggregation server with WebSocket round notifications, weighted delta aggregation, and a full client SDK supporting submit, pull, and automated train-loop workflows across distributed clients.
- Shipped production-grade security including Gaussian differential privacy, Byzantine defenses (L2 norm bounding and z-score outlier detection), Bearer token auth, and safetensors-only serialization — validated by 165 tests.

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, SQL, Bash.
- **Frontend & Backend:** React.js, Next.js, Redux, Spring Boot, FastAPI, Django, Flask, Express.js, Node.js, REST APIs, GraphQL, OAuth 2.0, JWT.
- **Data & Streaming:** Apache Kafka, Apache Airflow, Apache Spark, PySpark, Pandas
- **Cloud & DevOps:** AWS (Lambda, Route 53), Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Terraform, Git
- **Databases:** MySQL, PostgreSQL, MongoDB, Oracle, MariaDB, DynamoDB, SQLite, IBM DB2
- **AI / ML:** LangChain, pgvector, RAG Pipelines, OpenAI API, XGBoost, Scikit-learn, MLflow, MCP (Model Context Protocol), Federated Learning, LlamaIndex, Vector Databases, Prompt Engineering, Agentic Workflows, Hugging Face Transformers
- **Monitoring & Testing:** Prometheus, Grafana, CloudWatch, Pytest, JUnit, Postman
- **Project Management & Methodologies:** Agile (Scrum), Waterfall, SDLC, Object-Oriented Programming (OOP), Code Splitting, Caching Strategies

EDUCATION

University of Florida | Master of Science in Computer Science

May 2024